4 In this question you must show detailed reasoning.

A curve has equation $y = x - 5 + \frac{1}{x-2}$. The curve is shown in Fig. 4.

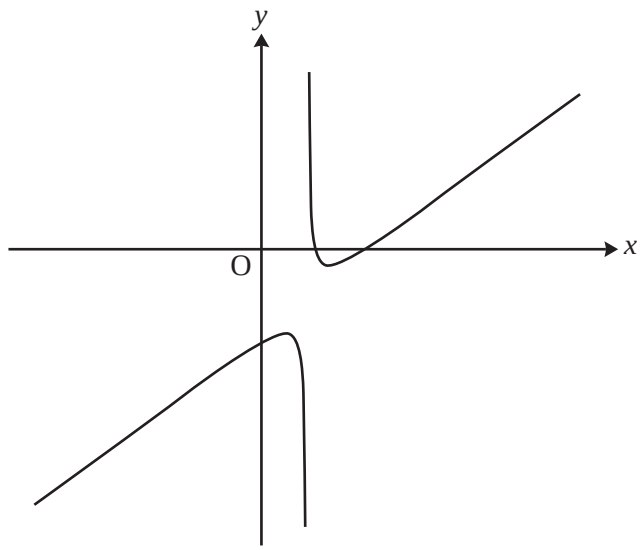

Fig. 4

(i) Determine the coordinates of the stationary points on the curve. **[5]**

(ii) Determine the nature of each stationary point. **[3]**

(iii) Write down the equation of the vertical asymptote. **[1]**

(iv) Deduce the set of values of x for which the curve is concave upwards. **[1]**